

Derek Lund

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EDUCATION

UCLA Samueli School of Engineering -- Expected June 2028

B.S. Aerospace Engineering

Orange Coast College -- August 2024 – May 2026

A.S.T. Aerospace / Mechanical Engineering

- Activities: Mu Alpha Theta (Director of Activities), Coast Aeronautical/Astronautical Research Lab (Aerodynamics Team Lead)
 - Relevant Coursework: Solid Fuel Propulsion, Differential Equations, MATLAB, Statics, Multivariable Calculus, Linear Algebra
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SKILLS

Software & Programming: ANSYS Fluent, OpenRocket, OnShape, Siemens NX, MATLAB, Python, PowerShell, Arduino, Excel

Aerospace: CFD Aerodynamic Analysis, wind tunnel & flight data analysis, rocket structures, avionics fundamentals, recovery systems

Methods: mathematical modeling, data analysis, scientific writing, LaTeX

EXPERIENCE

Aerodynamics Team Lead — IREC Competition | Coast Aeronautical/Astronautical Research Lab

November 2025 – Present

- Conducted comparative CFD studies in ANSYS Fluent to evaluate aerodynamic performance across multiple rocket geometries.
- Designed full-scale CAD models of selected configurations based on CFD-driven aerodynamic optimization results.
- Built modular 3D-printed subscale models for wind tunnel validation of simulated force and stability predictions.
- Analyzed aerodynamic forces and pitching moment trends across subsonic and transonic flight conditions.
- Developing a guided parachute recovery system and associated control logic to improve payload landing accuracy.
- Conducting CFD studies in ANSYS Fluent for recovery system to determine maximum loading and design requirements.

Independent Rocket Project

September 2025 – Present

- Built and flew a Level 1 high-power rocket, gaining hands-on experience with structural assembly and recovery systems.
- Created a full CAD assembly and simulated flight trajectories in OpenRocket to optimize recovery deployment timing.
- Applied aerodynamic stability principles and flight analysis methods to validate real-world vehicle performance.

Independent Mathematics Research

June 2025 – Present

- Coauthored a mathematical proof involving triangle intersections, structured through four independent analytical lemmas.
- Investigated graph theory applications related to transistor network layouts and data transfer optimization problems.

IT Intern | Mainfreight Inc.

June 2025 – August 2025

- Automated migration workflows for more than 1,000 users from Work Folders to OneDrive using PowerShell.
- Assisted with group policy administration and developed Excel-based tracking systems for project milestone management.

Research Presenter | Giles T. Brown Symposium

September 2024 – May 2025

- Modeled gravitational interactions in binary-body systems and generated MATLAB visualizations for technical presentation.
- Presented analytical findings to faculty evaluators while strengthening technical communication and research skills.

Accounting Intern | Mainfreight Inc.

June 2024 – August 2024

- Processed invoices within accounts receivable and accounts payable systems for the corporate finance department.
- Maintained financial and property tax documentation supporting organizational record management processes.

Independent Tutor

January 2024 – Present

- Provided individualized mathematics and science tutoring for middle school and high school students.
- Adapted instructional approaches for both online and in-person learning environments based on student needs.